Studies on the catalytic activity of hexokinase provided the following data:

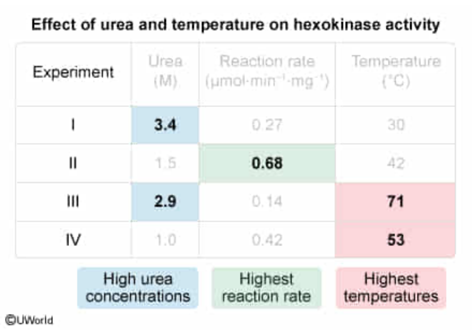
Experiment	Urea (M)	Reaction rate ($\mu mol \cdot min^{-1} \cdot mg^{-1}$)	Temperature (°C)
I	3.4	0.27	30
II	1.5	0.68	42
III	2.9	0.14	71
IV	1.0	0.42	53

Which experiment would contain the greatest percentage of folded protein?

- A. Experiment I [16%]
- B. Experiment II [59%]
- C. Experiment III [13%]
- D. Experiment IV [9%]

Correct
60% answered correctly

Explanation:


Effect of urea and temperature on hexokinase activity

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Proteins have evolved to function within a narrow range of cellular conditions. In particular, enzymes can remain functional only when their three-dimensional shape, or **tertiary structure**, is maintained. Noncovalent interactions, such as hydrogen bonds, ionic bonding, and hydrophobic effects, allow proteins to fold into secondary structures (eg, α-helices, β-sheets) that combine to form tertiary structures. The disruption of noncovalent bonds that causes proteins to unfold and lose their function is referred to as **denaturation**. Proteins unfold rapidly at conditions outside their optimal range because denaturation is a cooperative process.

Although heat energy increases the rate of an enzymatic reaction, **high temperatures** can also cause denaturation of the enzymes. The excess kinetic energy provided by increasing temperatures generates vibrations that disrupt hydrogen bonding between polar atoms and hydrophobic interactions in nonpolar regions. Denaturation also occurs in the presence of **urea**, which is a polar amide produced by protein metabolism and excreted by the body. Urea induces the unfolding and denaturation of proteins by exposing hydrophobic groups and interfering with hydrogen bonding in the amino acid backbone.

Proteins exposed to high temperatures or to high concentrations of urea **unfold easily** and **lose their function**. Therefore, high temperatures or high urea concentrations decrease the enzymatic reaction rate, indicating a smaller percentage of folded protein. Of the choices listed, Experiment II provides the best conditions for enzymatic function because the temperature and urea concentration used allow for the **highest enzymatic reaction rate**. Consequently, Experiment II must contain the highest percentage of **folded, functional protein**.

(Choices A, C, and D) These experiments all show lower enzymatic activity than Experiment II, indicating that a higher percentage of enzymes were no longer folded nor functional in these experiments. The enzymes in Experiments III and IV were likely denatured by high temperatures whereas the enzyme in Experiment I was likely denatured by the high urea concentration.

Educational objective:
Enzymatic function is maintained by noncovalent interactions in the tertiary and secondary structure of proteins. Noncovalent interactions such as hydrogen bonding and hydrophobic effects are disrupted by high temperatures and high concentrations of urea. Proteins lose functionality when their tertiary structure unfolds (denaturation).

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Subject:
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Amino Acids and Proteins